

Remote Sensing Subject-Matter Primer

Satellite Imagery, Commercial Providers, Change Detection, and Object Detection

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How To Read This Primer

This primer is meant to be used in three passes:

1. First pass: understand the sensor and market landscape.
2. Second pass: understand change detection and object detection workflows.
3. Third pass: connect those concepts into an evidence-bearing system like SHIELD.

If you only need the shortest possible study order, read these sections first:

- 1. Remote Sensing Fundamentals
- 2. Main Satellite Sensor Types
- 7. How Imagery Companies Usually Work
- 9. Change Detection
- 10. Object Detection in Overhead Imagery
- 15. What SHIELD Needs to Understand from All This

Purpose

This document is a subject-matter primer, not a product brief.

It is meant to teach the technical concepts that a team needs to understand before it can build serious capabilities around:

- aircraft taxonomy from overhead imagery
- resolution-aware image analysis
- satellite and SAR imagery sourcing
- change detection over time
- object detection and fine-grained recognition
- evidence-bearing workflows inside a system like SHIELD

It is deliberately broader than MIAR. The goal is to understand the domain itself:

- how the Earth observation market works
- what kinds of imagery exist
- what commercial and public providers actually offer today
- how imagery is consumed in real workflows
- what modern analysis pipelines do well
- where the hard limitations still are

Executive View

If you are new to this domain, the first thing to understand is that “satellite imagery” is not one thing.

It is an ecosystem built from different tradeoffs:

- optical vs radar
- very high spatial resolution vs broad area coverage
- daily revisit vs exquisite detail
- open public data vs premium commercial tasking
- single-scene interpretation vs long-running time-series monitoring

Most failed geospatial AI efforts make one of four mistakes:

- 1. They treat resolution as the only thing that matters.
- 2. They ignore how imagery is actually purchased, tasked, licensed, and delivered.
- 3. They underestimate change detection and time-series analysis.
- 4. They confuse object detection demos with operational evidence workflows.

The right mental model is:

- 1. learn the sensor stack
- 2. learn the supplier landscape
- 3. learn the data delivery and consumption models
- 4. learn the analysis patterns
- 5. learn the failure modes
- 6. only then decide what product or model program to build

At A Glance

Topic	What you need to know first	Why it matters
Resolution estimate what a prove	Spatial resolution is only one of four resolution types	Many teams overe nominal GSD can
Sensors stacks usually	Optical and SAR solve different problems	Real monitoring need both
Providers ence workflows agery is sold	Archive, tasking, monitoring, licensing, and APIs matter as much as pixels	Product and evid depend on how im and delivered
Change detection one of the failure modes	Registration and normalization are as important as model choice	False change is core operational
Object detection evidence in real	Benchmark mAP is not the same as operational success	Domain shift and quality dominate deployments
SHIELD alignment rovenance, nd governance	Detections should enter as evidence first, not canonical truth	This preserves p reviewability, a

1. Remote Sensing Fundamentals

1.1 What remote sensing means

Remote sensing is the measurement or observation of the Earth from a distance. In this context, that usually means:

- satellites
- aircraft
- drones

For this primer, the primary focus is Earth observation satellites.

1.2 The four resolution concepts everyone must know

When people say “resolution”, they often mean only spatial resolution. That is too narrow.

There are four distinct ideas:

Spatial resolution

The ground size represented by one pixel.

Examples:

- 30 cm
- 50 cm
- 1 m
- 10 m
- 30 m

This determines how much geometric detail can be resolved.

Temporal resolution

How often the same place can be imaged again.

Examples:

- several times per day in high-demand commercial constellations
- near-daily land coverage with medium-resolution fleets
- every few days with paired public missions

This determines whether you can monitor change or only inspect snapshots.

Spectral resolution

How many wavelength bands are measured, and how narrow those bands are.

Examples:

- RGB only
- RGB + near-infrared
- multispectral
- hyperspectral

This determines whether you can do more than shape-based visual interpretation.

Radiometric resolution

How finely the sensor records intensity differences.

This affects subtle discrimination, calibration quality, and analytic usefulness

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1.3 Ground sample distance is important, but not enough

Ground sample distance, or GSD, is the most widely quoted number in the market because it is easy to compare across providers. But GSD is not the full story.

Interpretability also depends on:

- look angle or off-nadir collection
- atmospheric haze
- cloud cover
- shadows
- geolocation accuracy
- image processing pipeline
- pansharpening quality
- sensor noise
- compression
- whether the target is in the open or obscured

A nominal 30 cm image can still be poor for recognition if the acquisition geometry or scene conditions are bad.

1.4 The operational distinction between seeing and knowing

In remote sensing, there is a major difference between:

- seeing something in the image
- proving what it is

This difference matters for all downstream tasks:

- aircraft family recognition
- force-counting
- change detection
- intelligence reporting

A useful rule:

- detection asks “is there an object here”
- classification asks “what object is it”
- interpretation asks “what does it mean”

Many teams are good at the first and weak at the third.

Quick reference: four kinds of resolution

Resolution type	Core question	Typical example
Spatial m	How much ground does one pixel cover?	30 cm, 50 cm, 10 m
Temporal -day	How often can I see the same place again?	daily, 5-day, 16-day

Spectral al,	Which wavelength bands are measured?	RGB, multispectr hyperspectral
Radiometric ne	How finely are intensity differences captured?	bit-depth and to discrimination
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2. Main Satellite Sensor Types

2.1 Optical electro-optical imagery

This is the imagery most people imagine first: photos of the Earth in visible light, often with near-infrared or other bands.

Strengths:

- intuitive for human interpretation
- excellent for object shape and scene context
- strongest ecosystem for conventional computer vision
- useful for maps, basemaps, site monitoring, and object recognition

Weaknesses:

- blocked by clouds
- dependent on daylight
- can be degraded by haze and shadow

For aircraft analysis, optical imagery is usually the primary modality for morphology and fine-grained recognition.

2.2 Multispectral imagery

Multispectral imagery captures a limited number of distinct wavelength bands, of ten in visible, near-infrared, and shortwave infrared.

Why it matters:

- vegetation and water separation
- material and land-cover cues
- cloud and shadow masking
- change detection robustness
- harmonized time-series analysis

For example, Sentinel-2 provides 13 spectral bands at multiple spatial resolutions and is used heavily for land and change applications.

2.3 Hyperspectral imagery

Hyperspectral imagery measures many narrow bands and is useful for material identification, anomaly detection, and some industrial or environmental tasks.

It is strategically important, but for military aircraft detection from overhead imagery it is not usually the first practical wedge. It is more relevant when:

- material discrimination matters
- plume or gas analysis matters
- advanced environmental signatures matter

2.4 Synthetic aperture radar

SAR is one of the most important concepts to learn if you want a serious monitoring capability.

Radar satellites illuminate the Earth actively and can image:

- at night
- through clouds
- in many weather conditions

Strengths:

- all-weather, day-night collection
- highly useful for repeated monitoring
- often excellent for change detection
- can support coherence and interferometric analysis

Weaknesses:

- much less intuitive for human analysts unfamiliar with radar phenomenology
- speckle, layover, and geometry can complicate interpretation
- object classification often requires special model design and domain expertise

For persistent military monitoring, SAR is not optional knowledge. Even if the end-user prefers optical imagery, the monitoring stack should understand SAR.

2.5 Thermal and other modalities

Thermal imagery can help with:

- fires
- heat signatures
- night analysis
- some industrial monitoring

But for most overhead aircraft recognition workflows, thermal is secondary relative to optical and SAR.

Quick reference: sensor types

Sensor type	Best at	Main limitation
Optical / E0 ht dependence	Human-readable scene interpretation and shape-based recognition	Cloud and daylight
Multispectral iled enough	Time series, land-cover, masking, environmental context	Usually not detailed for small-object

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Hyperspectral
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Material discrimination and
advanced signature analysis

Less common oper
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aircraft work

SAR
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All-weather, day-night
monitoring and change
detection

Harder human int
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requirements

Thermal
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Heat-related phenomena and
some night use cases

Usually secondar
aircraft recogni

3. Public Earth Observation Systems You Must Know

These systems are foundational because they anchor a huge amount of research, analytics, and operational preprocessing.

3.1 Sentinel-2

Sentinel-2 is the canonical free optical multispectral land mission for many workflows.

Key facts from Copernicus:

- 13 spectral bands
- 4 bands at 10 m
- 6 bands at 20 m
- 3 bands at 60 m
- 290 km swath
- paired mission with 5-day revisit in the combined constellation

What it is good for:

- land-cover and land-use analysis
- broad site context
- environmental and infrastructure monitoring
- medium-resolution change detection
- building time-series baselines

What it is not good for:

- fine-grained aircraft subtype recognition
- tactical vehicle identification
- small-object discrimination at military-site level

Sentinel-2 is essential for context and time series, but not sufficient for detailed aircraft recognition.

3.2 Sentinel-1

Sentinel-1 is the core public SAR mission to know.

Key facts from ESA and Copernicus:

- C-band SAR
- all-weather, day-night imaging
- resolution down to about 5 m in some modes
- coverage up to 400 km depending on mode
- historically designed around two-satellite operations, with newer spacecraft entering service after Sentinel-1B loss

What it is good for:

- broad persistent monitoring
- flood and maritime applications
- change detection
- coherence-based analyses
- weather-independent collection

What it is not good for:

- visually intuitive fine-grained aircraft recognition

It is strategically important because it teaches the team how monitoring behaves when optical is unavailable.

3.3 Landsat

Landsat is the historical backbone of moderate-resolution Earth observation.

USGS notes:

- continuous archive since 1972
- current Landsat 8 and 9 provide 30 m imagery
- each satellite revisits globally every 16 days, and together every 8 days

Why it matters:

- long historical time series
- calibration and scientific baselines
- land change studies
- harmonization with Sentinel-based products

Landsat is rarely the main source for military aircraft monitoring, but it is vital for time-series Earth observation literacy.

Quick reference: public systems

System	Modality	Typical scale of use	Best use
Sentinel-2 ext, time road change	Optical multispectral	10 m to 60 m bands	Land cont series, b
Sentinel-1 ndependent	C-band SAR	meter-class SAR modes	Weather-i

g and change

monitorin

Landsat 8/9
baseline

Optical multispectral 30 m

Historica
and long

time series

4. Commercial Optical Imagery Providers You Should Understand

This is not an exhaustive market map. It is a practical first-cut of the providers that shape current workflows.

4.1 Maxar / Vantor

Maxar's imagery business now appears under the Vantor brand in public web properties, but the core idea remains the same: premium, very high-resolution imagery with deep archive and tasking capabilities.

Current public claims include:

- same-location revisit up to 15x per day
- about 3.5 million sq km per day at 30 cm resolution
- integration of complementary sensors including Umbra SAR

Why this matters:

- premium optical imagery for detailed monitoring
- strong archive and tasking posture
- widely used for defense, mapping, and analytics

Operational takeaway:

This is the kind of provider teams use when detail matters and cost is acceptable.

4.2 Airbus

Airbus is one of the key premium competitors in optical and radar access.

Relevant current public material highlights:

- Pléiades Neo 30 cm imagery
- OneAtlas access to archive and tasking
- optical plus radar services in one platform
- reactive tasking and long archive access

Why it matters:

- strong European alternative in premium imagery
- good archive-plus-tasking model
- useful for algorithm training and multi-provider sourcing

4.3 BlackSky

BlackSky differentiates around speed, revisit, and integrated analytics.

Its current public material highlights:

- 35 cm Gen-3 imagery
- general availability as of March 12, 2026
- AI-enabled analytics in the BlackSky Spectra platform
- day, twilight, and nighttime collection positioning in its product story

Why it matters:

- monitoring-oriented posture
- stronger emphasis on timely insight products
- useful reference for customers who value alerting and persistent watch more than archive depth alone

4.4 Planet high-resolution optical

Planet’s high-resolution tasking side is anchored by SkySat and newer tasking products.

Planet’s current documentation says SkySat:

- consists of 15 satellites
- can revisit a location up to 10x daily
- produces 4-band and panchromatic imagery
- is sampled at 50 cm when orthorectified

Why it matters:

- strong tasking and API-first delivery model
- more accessible operational workflows for high-frequency collection
- useful midpoint between premium detail and programmatic access

Quick reference: commercial optical providers

Provider	Publicly positioned strength	What to remember
Maxar / Vantor and archive	Premium very-high-resolution imagery, archive, tasking	Use when detail depth matter
Airbus commercial supplier and tasking	Premium optical plus radar access through OneAtlas	Strong alternative model with archive
BlackSky commercial	Fast revisit and integrated analytics posture	Monitoring-oriented product story
Planet high-res / SkySat cadence	Frequent tasking and API-first workflows	Useful for high-frequency operational access

5. Commercial Medium-Resolution Providers and Daily Monitoring

5.1 PlanetScope

PlanetScope is the canonical example of a high-temporal, medium-resolution commercial system.

Planet documentation states:

- near-daily multispectral coverage over landmasses
- analysis-ready products at 3 m
- harmonized stacks designed for time-series analysis and machine learning

Why it matters:

- not good for aircraft subtype work
- excellent for persistent land monitoring, pattern-of-life at larger scales, and time-series analytics
- teaches an important lesson: cadence can matter more than exquisite detail

This is a central concept for change detection. A daily 3 m feed is often more valuable for trend monitoring than occasional premium 30 cm images.

6. Commercial SAR Providers You Should Understand

6.1 ICEYE

ICEYE is one of the most important commercial SAR providers to know for monitoring applications.

Current public product messaging emphasizes:

- SAR designed for change detection
- imaging modes from wide-area monitoring to very high resolution
- ground resolutions including 25 cm, 50 cm, and 1 m in dwell modes

Why it matters:

- strong fit for persistent monitoring
- useful when cloud cover makes optical unreliable
- emphasizes the reality that radar is often the monitoring backbone rather than a niche

6.2 Capella Space

Capella is a major commercial SAR provider and strong reference point for U.S.-based SAR tasking and archive access.

Current public material highlights:

- 50 cm x 50 cm Spotlight imagery
- archive access plus tasking
- SAR delivery through cloud-oriented workflows
- time-sensitive tasking features and archive search

Why it matters:

- practical operational SAR access
- strong reference for how modern SAR companies expose tasking and archive

- useful for customers who need radar without building a full radar program themselves

6.3 Umbra

Umbra is important because it pushes very high-resolution commercial SAR.

Public material highlights:

- 16 cm resolution image released in 2023
- complex data access better than 25 cm for some users and use cases

Why it matters:

- demonstrates how far commercial SAR has advanced
- raises the ceiling for all-weather site detail
- useful benchmark when discussing what “commercial radar” can now mean

6.4 The key SAR lesson

For serious monitoring, the question is not optical or SAR.

The question is:

- what part of the workflow depends on optical interpretation
- what part needs weather-independent revisit
- what part needs both

The strongest operational systems fuse both.

Quick reference: commercial SAR providers

Provider	Publicly positioned strength	What to remember
ICEYE er monitoring	Monitoring-oriented SAR and change-detection fit	Strong all-weath reference
Capella cess model for	High-resolution commercial SAR with tasking and archive	Practical SAR ac many customers
Umbra g of modern	Very-high-resolution SAR positioning	Shows the ceilin commercial SAR

7. How Imagery Companies Usually Work

This is the part many first-time builders overlook.

You are not just buying pixels. You are buying access patterns, rights, latency, and operational behavior.

7.1 The main acquisition models

Most providers expose some combination of:

- archive access
- new tasking
- subscriptions or repeat monitoring
- basemaps or mosaics
- API access
- cloud delivery

7.2 Archive imagery

Archive means imagery already collected in the past.

Why customers use it:

- historical backfill
- comparison against prior activity
- cheaper access than fresh tasking in many cases
- training and evaluation

Examples from current vendor documentation:

- Planet supports archive orders for high-resolution collects
- Airbus OneAtlas exposes archive ordering
- Capella exposes archive search and archive roles

Archive is fundamental for change detection and site history.

7.3 Tasking

Tasking means asking a provider to collect a new image over an AOI within a time window.

What tasking usually exposes:

- AOI geometry
- acquisition window
- quality constraints
- look-angle or off-nadir constraints
- cloud preferences
- priority tiers
- product type

This is where the business model changes from passive search to active collection.

7.4 Repeat tasking and monitoring

For defense and infrastructure workflows, repeat collection is often more important than a single exquisite scene.

Monitoring products typically allow:

- daily or periodic checks
- recurring AOIs
- event-driven collections
- alerting when new imagery arrives

This is operationally closer to “feed behavior” than to one-time image ordering.

7.5 Basemaps and mosaics

A basemap is not a scene. It is a processed, harmonized image layer built from many collections.

Use cases:

- visualization
- map context
- large-area feature extraction
- broad planning

Limits:

- can hide acquisition specifics
- may not retain scene-level provenance needed for evidence workflows

Basemaps are excellent context layers but not a substitute for scene-level evidence.

7.6 Delivery formats and access patterns

Modern imagery is increasingly delivered through:

- APIs
- cloud object storage
- streamable basemap endpoints
- cloud-optimized GeoTIFFs
- STAC-like catalog workflows

Operational systems should assume programmatic delivery, not manual download only.

7.7 Licensing and exclusivity

Commercial imagery usually comes with licensing terms that matter operationally:

- who may use the imagery
- whether it can be shared with partners
- whether derived outputs can be redistributed
- whether tasked imagery enters a shared archive after a delay

For example, Planet documents a tasking-to-archive model where high-resolution tasking is normally published to the archive by default, with some archive-withhold options available depending on contract.

This matters for sovereignty, secrecy, and downstream redistribution.

Quick reference: imagery business models

Model	What it means	Why it matters
Archive kfill and sis	Past collections already available	Required for back historical analysis
Tasking	New collection over a	Required for timely

e-sensitive	specified AOI and window	collection
Monitoring / repeat tasking checkbox behavior	Repeated or subscription-style	Required for wat
Basemaps / mosaics xt, weak for	Harmonized context layers	Useful for conte
ence	built from many scenes	scene-level evid
API / cloud delivery lable	Programmatic access to imagery and metadata	Required for sca pipelines

8. What “Consumption” Really Means in Practice

When teams ask how customers will “consume” imagery, the answer usually spans five patterns.

8.1 Visual inspection

Analysts look at scenes, zoom, compare, annotate, and build assessments.

8.2 GIS integration

Imagery becomes a layer in:

- desktop GIS
- web GIS
- map-based intelligence workspaces

8.3 Time-series analytics

The imagery is processed as a sequence rather than a standalone picture.

8.4 Model inference

The imagery is chipped, tiled, or streamed into detection, segmentation, or change models.

8.5 Evidence workflows

The resulting detections or changes are stored with:

- scene ids
- timestamps
- footprints
- chips
- analyst notes
- review state

This fifth pattern is the one most relevant to SHIELD-like systems.

9. Change Detection: What It Is and Why It Matters

9.1 The core definition

Change detection is the identification of meaningful differences between observa

tions from
different times.

That sounds simple, but operationally it is difficult because you must separate:

- real change on the ground
- apparent change from sensor or geometry differences

9.2 Types of change detection

There are several major types:

Pixel-level change detection

Finds changed pixels or regions directly.

Good for:

- land-cover shifts
- broad damage or flood mapping

Less ideal for:

- crisp object-level military interpretation

Object-level change detection

Compares detected objects or derived entities over time.

Good for:

- aircraft count change
- vehicle appearance/disappearance
- new structures

This is often the most operationally useful form in defense monitoring.

Semantic change detection

Not only says that something changed, but what kind of change happened.

Examples:

- empty apron to occupied apron
- civilian aircraft replaced by military aircraft
- open terrain to new revetment construction

Time-series or continuous change detection

Works over longer sequences rather than just two dates.

This matters because many defense questions are not binary before/after questions. They are pattern questions:

- is activity increasing
- is dispersal becoming more common
- is a site preparing for a surge

9.3 Why change detection is hard

False changes are caused by:

- registration error
- seasonal variation
- different illumination
- cloud and shadow
- off-nadir changes
- atmospheric effects
- different sensor modalities
- SAR speckle and geometry effects

In practice, most of the engineering work is about controlling these sources of false change.

9.4 Preprocessing is not optional

Before doing serious change detection, you typically need:

- geometric co-registration
- radiometric normalization
- cloud and shadow masks
- sensor harmonization
- valid-pixel masks

Planet’s analysis-ready products are instructive here because they exist precisely to improve temporal consistency for time-series and machine learning uses.

9.5 What is out there today

The current state of the art is strongly deep-learning oriented, but the field is still mixed.

Broad buckets:

- classical image differencing and thresholding
- feature- or index-based methods
- Siamese CNN approaches
- transformer-based change detection networks
- hybrid CNN-transformer models
- semantic change detection models

Recent review literature shows:

- the field is advancing quickly
- open datasets matter a lot
- supervision remains expensive
- multi-sensor and cross-domain change remain hard

The important strategic takeaway is that change detection is not a solved commodity when you move into operational, multi-source, mixed-quality military imagery.

Quick reference: change detection task types

Type	Best for	Limitation
Pixel-level	Broad surface change	Weak semantic pr

ecision for

Object-level
ble detection

Semantic
d stronger

Time-series / continuous
r temporal
storage

Count change and
appearance/disappearance

Meaningful typed change

Trend and pattern-of-life
analysis

military objects

Depends on relia
first

Harder labels an
models required

Requires stronge
consistency and
discipline

10. Object Detection in Overhead Imagery

10.1 What it is

Object detection localizes and labels objects in the image, usually with:

- bounding boxes
- oriented boxes
- polygons or masks

For overhead imagery, this differs from everyday computer vision because:

- targets are small relative to the scene
- orientations vary widely
- classes can be visually similar
- context differs from natural-image benchmarks

10.2 Why overhead detection is special

Satellite images are large and sparse.

A scene may contain:

- huge empty areas
- cluttered infrastructure
- highly imbalanced class frequencies
- targets only a few dozen pixels across

This changes model design and evaluation significantly.

10.3 Main task types

You should distinguish:

- detection
- classification
- fine-grained classification
- segmentation
- tracking across time

These are related but not interchangeable.

For military aviation:

- detection is finding aircraft
- classification is broad role or family
- fine-grained classification is subtype or variant
- tracking across time is appearance and disappearance across scenes

10.4 Model families commonly used today

In practice, the detection ecosystem is built from a few broad model families:

- one-stage detectors such as YOLO variants
- two-stage detectors such as Faster R-CNN
- anchor-free detectors
- transformer-based detectors such as DETR families
- oriented object detectors for rotated targets

10.5 What current research says

A 2024 comparative review focused on aircraft detection in satellite imagery found strong performance from YOLO-family models in that benchmark setting, with YOLOv5 leading the evaluated methods on the tested datasets.

Do not over-generalize that finding into:

- all datasets
- all resolutions
- all deployment constraints

But it is a good practical signal:

- modern detector choice still matters
- benchmark success is measurable
- aircraft detection is increasingly tractable on curated datasets

10.6 The hard part is not just boxes

Operational difficulty comes from:

- domain shift across providers
- limited labels for rare military classes
- confusion among related airframes
- low-quality or off-angle scenes
- shelters, camouflage, and clutter
- need for calibrated confidence rather than raw scores

This is why product teams should not confuse good benchmark mAP with operational success.

Quick reference: analysis stack

Task	Question answered	Typical output
Detection	Is there an object here?	box, mask, score
Classification	What broad thing is it?	role or family label

Fine-grained recognition is	Which subtype or variant is it?	subtype hypotheses
Change detection, or	What changed between dates?	delta object, resemantic change
Interpretation assessment	What does it mean operationally?	analyst claim or

11. Fine-Grained Recognition: The Next Difficulty Step

Detection is often easier than subtype recognition.

For aircraft specifically, the challenge increases because:

- many variants differ by small shape cues
- overhead view removes side-profile details
- wing and tail geometry can be ambiguous
- shadows and orientation distort visual evidence

A disciplined system therefore needs hierarchical outputs:

- object present
- broad role
- family
- subtype if supported

This is one of the most important conceptual lessons in the domain.

12. Data and Benchmarks You Should Know

A few public benchmarks are especially useful for orientation:

xView

The DIU xView challenge dataset is one of the best-known overhead object detection benchmarks and includes over one million objects across 60 classes in large-scale imagery.

Why it matters:

- strong baseline overhead-detection reference
- broad object classes
- useful for detection literacy

RarePlanes

RarePlanes was designed to study aircraft detection and the role of synthetic data for overhead imagery.

Why it matters:

- aircraft-focused
- mixes real and synthetic thinking
- relevant to low-data and military-style scenarios

FAIR1M

FAIR1M is a large fine-grained remote sensing benchmark with more than a million instances.

Why it matters:

- fine-grained recognition
- oriented object detection
- category richness beyond coarse object labels

The benchmark lesson

Public datasets are essential for experimentation, but they are not the same thing as operational truth.

They often differ from customer conditions in:

- geography
- provider mix
- resolution mix
- class priors
- cloud and weather behavior
- licensing and reproducibility

13. Foundation Models and What They Mean Here

Foundation models are now part of the remote sensing landscape, but their role should be understood carefully.

13.1 What they are

These are large pre-trained models trained on huge remote-sensing corpora and then fine-tuned for downstream tasks.

13.2 Relevant current example

NASA's Prithvi geospatial foundation model is one of the most important public examples.

NASA's 2026 public material notes:

- on-orbit deployment of Prithvi as the first geospatial foundation model demonstrated in orbit
- training on long historical Earth observation data
- use across multiple Earth observation tasks

NASA technical reporting on Prithvi-E0-2.0 also reports:

- training on 4.2 million global time-series samples from harmonized Landsat and Sentinel-2 data
- benchmarking gains over earlier models and several other geospatial foundation models

13.3 What this means practically

Foundation models matter because they can:

- reduce labeled-data needs
- transfer across tasks
- improve representation learning
- support multi-temporal reasoning

But they do not remove the need for:

- domain-specific labels
- target ontology
- rigorous evaluation
- deployment engineering

For high-resolution military object localization, task-specific fine-tuning still matters a lot.

14. Common Failure Modes in Real Systems

These are the mistakes teams repeatedly make.

14.1 Resolution overconfidence

They assume a nominal GSD guarantees target identity.

14.2 Archive blindness

They think only about new tasking and forget that historical analysis depends on archive depth.

14.3 Optical-only thinking

They ignore SAR until weather or revisit constraints force them to care.

14.4 Benchmark overfitting

They optimize for public datasets and then fail on provider, geometry, or geography shift.

14.5 Ignoring evidence lineage

They store detections without:

- scene ids
- acquisition timestamps
- model versions
- quality metadata

That makes later review weak.

14.6 Confusing basemaps with evidentiary scenes

Basemaps are excellent context, but they often strip away the exact acquisition identity needed for defensible analysis.

15. What SHIELD Needs to Understand from All This

If these subject matters are to connect well with SHIELD, the critical architectural

boundary is simple:

- scenes, detections, chips, and change products belong first in the evidence plane
- reviewed interpretations can become claims
- only governed, analyst-approved conclusions should influence canonical ORBAT & ruth

That means SHIELD should treat the remote-sensing stack as:

- imagery source and scene registry
- geospatial evidence layer
- annotation and claim environment
- review and promotion workflow

Not just as a map viewer.

15.1 Recommended internal objects

A SHIELD-aligned system should likely distinguish:

- source asset
- scene
- collection footprint
- derived chip
- detection
- compare result
- analyst annotation
- claim
- promotion decision

This makes the domain teachable and auditable.

15.2 Why this matters

The remote-sensing problem is not only model quality.

It is also:

- source management
- time alignment
- reviewability
- provenance
- evidence promotion

Those are exactly the places where a product either becomes serious or remains a demo.

16. A Study Order for the Team

If you want a disciplined learning sequence, use this order:

1. Learn optical, multispectral, and SAR differences.
2. Learn spatial, temporal, spectral, and radiometric resolution.
3. Learn the public systems first: Sentinel-1, Sentinel-2, Landsat.
4. Learn the commercial provider models: archive, tasking, repeat monitoring, base maps.
5. Learn preprocessing and harmonization.
6. Learn change detection before fine-grained object recognition.
7. Learn object detection and hierarchical classification.
8. Learn evidence and provenance design.
9. Only then choose model and product priorities.

17. Final Takeaway

To work well in this domain, a team needs to stop thinking of “satellite imagery AI” as a single capability.

It is really a stack:

- sensor understanding
- vendor and data-access understanding
- preprocessing and quality control
- temporal analysis
- object detection
- confidence management
- evidence workflow

The most valuable mindset shift is this:

the operational asset is not just the model.

It is the system that knows:

- what imagery exists
- what kind it is
- what it can and cannot support
- what changed
- what the model thinks
- what an analyst accepted
- and how all of that is traceable

That is the actual subject matter.

Appendix: One-Page Memory Frame

If you remember only one page from this primer, remember this:

1. 30 cm imagery is not the same thing as certainty.
2. Optical explains shape well; SAR explains persistence and all-weather monitoring.
3. Archive plus tasking matters more than one-off imagery access.
4. Time series and change detection are often more valuable than a single exquisite image.
5. Benchmark detection is easier than operational classification.
6. Detections are not intelligence truth until they are reviewed and linked to evidence.

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